Explain and evaluate multi-step solutions

Use the scaffold, show each step and ask for an adult check when marked.

Name: ____ Date: ________________

1. Read the model: Two methods solve 48 x 125 - 3,760. One multiplies first; one subtracts first. Which is valid? Copy or complete the first step.

2. A solution to 3,840 / 24 x 15 gives 10. Find the error and correct it.

3. Solve 96 x 45 + 8,640 / 32.

4. Miro says: Miro assumes every number in a word problem must be used. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: Two methods solve 48 x 125 - 3,760. One multiplies first; one subtracts first. Which is valid? Copy or complete the first step.

   Answer Multiplication must represent the total books first, so multiply first. The answer is 2,240.

2. A solution to 3,840 / 24 x 15 gives 10. Find the error and correct it.

   Answer 3,840 / 24 = 160, then 160 x 15 = 2,400.

3. Solve 96 x 45 + 8,640 / 32.

   Answer 4,590.

4. Miro says: Miro assumes every number in a word problem must be used. Is this correct? Explain.

   Answer Use only information that changes the quantity asked for.

Explain and evaluate multi-step solutions

Work independently. Show a complete method and check at least one answer.

Name: ____ Date: ________________

1. A solution to 3,840 / 24 x 15 gives 10. Find the error and correct it.

2. Solve 96 x 45 + 8,640 / 32.

3. Write a multi-step problem with an unnecessary piece of information and explain why it is unnecessary.

4. Identify and correct this misconception: Miro assumes every number in a word problem must be used.

5. Write a complete sentence explaining how you checked one answer.

Teacher answers and checking prompts

1. A solution to 3,840 / 24 x 15 gives 10. Find the error and correct it.
   Answer 3,840 / 24 = 160, then 160 x 15 = 2,400.

2. Solve 96 x 45 + 8,640 / 32.
   Answer 4,590.

3. Write a multi-step problem with an unnecessary piece of information and explain why it is unnecessary.
   Answer Answers vary. The unused information must not affect the required calculation.

4. Identify and correct this misconception: Miro assumes every number in a word problem must be used.
   Answer Use only information that changes the quantity asked for.

5. Write a complete sentence explaining how you checked one answer.
   Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.

Explain and evaluate multi-step solutions

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: ____ Date: __________________

1. Write a multi-step problem with an unnecessary piece of information and explain why it is unnecessary.

2. Explain why this reasoning fails: Miro assumes every number in a word problem must be used.

3. Create a new example that tests the same idea as: Solve 96 x 45 + 8,640 / 32.

4. Find a second method or representation. Compare its efficiency with your first method.

5. Write one always, sometimes or never statement about today's learning and justify it.

Teacher answers and checking prompts

1. Write a multi-step problem with an unnecessary piece of information and explain why it is unnecessary.
   Answer Answers vary. The unused information must not affect the required calculation.

2. Explain why this reasoning fails: Miro assumes every number in a word problem must be used.
   Answer Use only information that changes the quantity asked for.

3. Create a new example that tests the same idea as: Solve 96 x 45 + 8,640 / 32.
   Answer Answers vary. The example and solution must preserve the mathematical structure.

4. Find a second method or representation. Compare its efficiency with your first method.
   Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.

5. Write one always, sometimes or never statement about today's learning and justify it.
   Answer Answers vary. A valid example and counterexample should support the classification.